

CONTRACTOR TECHNICAL SERIES

Diamond Tool Field Guide

Select, mount, operate, troubleshoot, and reorder professional diamond tooling with less guesswork.

THE SIX FACTS THAT DRIVE EVERY RECOMMENDATION

1. SAW	2. MATERIAL	3. AGGREGATE
Model, horsepower, arbor	Concrete age, asphalt, masonry, stone	Hard, medium, soft, unknown
4. CUT	5. COOLING	6. PRODUCTION
Diameter, depth, rebar	Wet, dry, available flow	Daily footage, speed, finish

Use the Blade Finder at tdusales.com/blade-finder or call (480) 470-2577 when any one of these facts is unknown.

Safety first

- Use only a blade specifically rated for the machine, material, direction, and operating speed.
- Never exceed the lower of the blade's marked maximum RPM or the saw manufacturer's limit.
- Inspect guards, flanges, arbor fit, blade condition, water delivery, and rotation before every shift.
- Follow the saw manual, blade label, applicable ANSI/OSHA requirements, and the jobsite safety plan.

BLADE SELECTION

Match the bond to the material

Diamond crystals do the cutting while the bond controls how quickly worn diamonds release. The correct bond exposes fresh cutting points at a rate matched to the material's abrasiveness.

Material condition	Typical selection direction	Watch for
Hard cured concrete / hard aggregate	Softer bond	Glazing, slow cutting, heat
Green concrete / asphalt / abrasive block	Harder bond with undercut protection	Fast segment wear, core undercut
Reinforced concrete	Application-rated blade and suitable horsepower	Heat, pinching, side pressure
Tile / porcelain / glass	Continuous or application-specific rim	Chipping, overheating, feed pressure
Granite / engineered stone / quartzite	Stone-specific bond and rim geometry	Deflection, edge quality, coolant flow

Segment and rim choices

- Segmented rims prioritize debris clearance and cooling for production cutting.
- Turbo rims balance speed with a cleaner edge on masonry, stone, and tile applications.
- Continuous rims prioritize finish quality for tile, glass, and delicate materials.
- Undercut protection helps protect the steel core where abrasive slurry attacks beneath the segment.

Wet or dry

Use water whenever the product and machine are designed for it. Adequate flow cools the segment, controls dust, and carries fines out of the cut. A dry-rated blade still needs cooling pauses and correct feed pressure; never force a blade that is slowing or changing sound.

OPERATING REFERENCE

Cutting depth and RPM

Depths are approximate. Flange diameter, saw geometry, blade diameter, and component condition change usable depth. High-speed saw limits can differ from general recommended speeds.

Never exceed the marked blade or saw limit.

Blade	Typical depth	Tile depth	Rec. RPM	Max RPM
4	1	3/4	9,075	15,000
4-1/2	1-1/4	1	8,065	13,300
5	1-1/2	1-1/4	7,255	12,000
7	2-1/2	2-1/4	5,185	8,730
8	3	2-3/4	4,535	7,640
10	3-3/4	3-3/4	3,630	6,115
12	3-5/8 flat	-	3,025	5,095
14	4-5/8 flat / 5 masonry	-	2,595	4,365
16	5-5/8 flat / 6 masonry	-	2,270	3,820
18	6-5/8 flat / 7 masonry	-	2,015	3,395
20	7-5/8 flat / 8 masonry	-	1,815	3,055
24	9-5/8 flat	-	1,510	2,550
30	11-3/4 flat	-	1,120	2,040
36	14-3/4 flat	-	1,010	1,700
42	17-3/4 flat	-	865	1,455
48	19-3/4 flat	-	755	1,275

SETUP & OPERATION

A clean mount makes a clean cut

Before mounting

- Disconnect power or disable the engine. Inspect the blade for cracks, missing segments, distortion, glazing, and shipping damage.
- Clean the arbor, flanges, and contact surfaces. Confirm arbor size, drive pin, rotation arrow, and flange diameter.
- Do not use makeshift bushings, damaged flanges, or a blade that does not seat flat.

During the cut

- Bring the blade to operating speed before entering the material and keep the cut straight.
- Use steady feed pressure. Do not twist, pry, side-load, or use a cutting blade for unauthorized grinding.
- Maintain specified water flow on wet systems and control silica dust with compliant engineering controls.
- Stop if vibration, unusual noise, color change, loss of tension, or segment damage appears.

After the cut

- Allow the blade to stop normally. Inspect it before the next operation and record unusual wear while the job conditions are still known.

Equipment	Information to confirm
Angle grinder	Guard, arbor, side handle, rated diameter/RPM
High-speed cutoff saw	Rotation, drive pin, guard, dry/wet rating, max RPM
Walk-behind saw	Horsepower, shaft speed, flange, depth, water flow
Masonry / tile saw	Capacity, arbor, cart alignment, pump flow, material support
Core rig	Thread, diameter, stand anchoring, water swivel, RPM/gear

TROUBLESHOOTING

Read the wear pattern

Symptom	Likely checks	Immediate response
Will not cut	Bond too hard, glazing, low power, wrong rotation	Stop forcing; verify application and approved dressing method
Short life	Bond too soft, abrasive material, low water, misalignment	Verify blade family, flow, bearings, and tracking
Segment loss	Heat, impact, pinching, loose mount, undercut	Stop immediately; remove blade from service
Cracked core	Side pressure, twisting, impact, loose flanges	Stop immediately; never repair or reuse
Uneven wear	Bad bearings, bent shaft, misalignment, unequal water	Service saw and correct water delivery
Excess vibration	Arbor fit, flanges, blade flatness, bearings, overspeed	Stop immediately and inspect complete system
Burning / discoloration	Insufficient cooling, excessive feed, wrong bond	Stop; correct cooling and application before reuse

Capture what happened

Record the SKU, blade diameter, saw model, RPM, material and aggregate, wet/dry condition, footage, depth, failure location, and clear photos. This turns a complaint into an actionable technical diagnosis.

CORE DRILLING

Control the rig before the bit

- Anchor or vacuum-mount the stand exactly as the equipment manufacturer requires; confirm the base cannot shift.
- Match bit diameter, barrel length, thread, segment specification, motor power, and gear/RPM to the material.
- Start square, establish a stable kerf, and use enough water to cool the segments and evacuate slurry.
- When rebar is encountered, maintain alignment and controlled feed. Do not lever the barrel or force a stalled motor.
- If the bit binds, stop power before attempting removal. Check loose aggregate, segment condition, core wedging, and stand movement.

Common drilling symptoms

Symptom	Check first
Slow penetration	Bond/application, glaze, RPM/gear, feed, motor load
Barrel binds	Stand movement, loose core, debris evacuation, bent barrel
Segment wear accelerates	Abrasive material, water flow, excessive RPM or pressure
Core breaks repeatedly	Rig alignment, rebar, interrupted cut, material condition
Water does not return	Blocked kerf, insufficient flow, deep-hole evacuation

JOB & REORDER RECORD

Make the next recommendation better

Field	Record
Customer / job	
Saw model / horsepower	
Material / aggregate	
Blade or bit SKU	
Diameter / arbor / thread	
Depth / rebar	
Wet or dry / water flow	
Daily footage	
Observed life / speed	
What should improve	
Recommended next configuration	

Titan Diamond application support

Call (480) 470-2577 or use the Blade Finder. Product availability, machine compatibility, and specifications must be confirmed before ordering.

Technical reference compiled by Titan Diamond USA from manufacturer data and general diamond-tool operating principles. It does not replace the product label, machine manual, qualified operator training, or applicable safety requirements.